

Predictive Storage Caching and Tiered Data Migration

Motivation

Modern ML workloads break traditional storage systems

- Deep learning pipelines have heterogeneous I/O patterns
 - Training → random access
 - Inference → sequential reads
 - Checkpointing → large writes
- Static caching policies (LRU, LFU) assume stationary behavior
- Result:
 - Cache thrashing
 - Poor tier utilization
 - High cost (over-provisioning HBM)

Can we *adapt storage decisions dynamically* based on workload phase?

Three Classes of Access Patterns

Random / Zipfian Access

- Skewed frequency distribution
- Few blocks accessed frequently
- Many blocks accessed rarely
- Strong temporal locality

Example: Training with shuffled datasets (e.g., ResNet)

2. Sequential Access

- Ordered, contiguous reads
- Minimal or no reuse
- Strong spatial locality

Example: Streaming datasets (e.g., BERT)

3. Large-Block / Streaming Access

- Accesses large chunks of data
- High I/O size per request
- Often used for writes or volumetric reads

Example: Checkpointing / 3D data (e.g., UNet3D)

Why these three?

- MLCommons (35+ organizations) selected exactly these three patterns for the MLPerf Storage benchmark
- Represent the three dominant real-world ML training scenarios: vision, language, volumetric/medical

Characterizing Machine Learning I/O with MLPerf Storage

Oana Balmau

CHEOPS @ EuroSys, May 8th, 2023



MLPerf-Calibrated Trace Generation

Workload	Pattern	File Size	Rows
ResNet50	Zipfian ($\alpha=1.2$)	150KB	500K
BERT	Sequential	4KB	500K
UNet3D	Random crop	500MB	200K

Model Design

Two-Stage ML Architecture

1. Phase Classifier (LSTM)

- Input: sequence of block IDs
- Output: phase prediction

2. Conditioned Policy (MLP)

- Input:
 - sequence window
 - predicted phase embedding
- Output:
 - tier placement
 - prefetch probability

Storage Hierarchy

4-Tier System

Tier	Device	Latency	Cost
0	HBM	100ns	\$30/GB
1	CXL DRAM	80ns	\$8/GB
2	NVMe SSD	100μs	\$0.30/GB
3	Cold Storage	1ms	\$0.03/GB

Benchmark Results

Workload	ML Policy	LRU	LFU
ResNet	79.62%	79.66%	78.98%
BERT	99.99%	0%	0%
UNet3D	99.93%	99.52%	99.52%

Sequential workloads destroy classical policies

- LRU / LFU $\rightarrow$ 0% hit rate (BERT)
- ML model:
 - detects phase
 - activates prefetch

Generalization Test

Model	Hit Rate (Mixed)
ML Policy (Ours)	75.43%
LRU	35.21%
LFU	40.71%

System Performance

- Inference latency: ~46 microseconds
- Works within real-time constraints
- Scales to large traces (~1.2M accesses)